Classifying Oxides

Exam code: 0620 & 0971

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Classifying oxides

What are oxides?

- Oxides are compounds made from one or more atoms of oxygen combined with one other element
- Examples of oxides include:
  - Magnesium oxide - MgO
  - Zinc oxide - ZnO
  - Potassium oxide - K_2O
  - Carbon dioxide - CO_2
  - Sulfur dioxide - SO_2
  - Water - H_2O
- Oxides can be classified based on their acid-base characteristics

Acidic oxides

- Acidic oxides are formed when a **non-metal** element combines with oxygen
- They react with bases to form a salt and water
- When dissolved in water they produce an **acidic** solution with a **low** pH
- Common examples include CO_2, SO_2, NO_2 and SiO_2

Basic oxides

- Basic oxides are formed when a **metal** element combines with oxygen
- They react with acids to form a salt and water
- When dissolved in water they produce a **basic** solution with a **high** pH
- Common examples include CuO and CaO

Basic and acidic oxides

Help me understand

Stuck on something?

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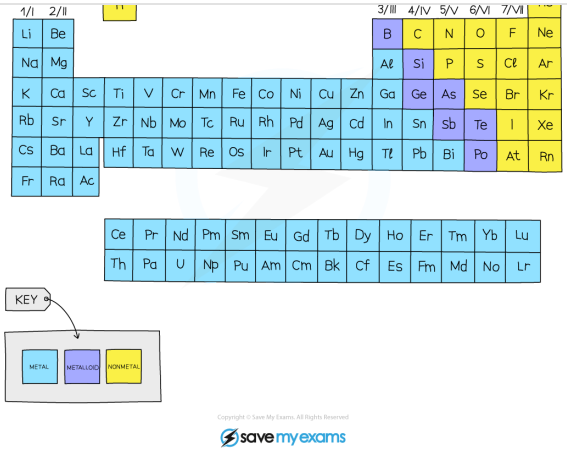

Metals form basic oxides while non-metals form acidic oxides

Amphoteric oxides

Extended tier only

Neutral oxides

- Some oxides do not react with either acids or bases and thus are said to be **neutral**
- Examples include N_2O, NO and CO

Amphoteric oxides

- Amphoteric oxides are a curious group of oxides that can behave as **both acidic** and **basic**, depending on whether the other reactant is an acid or a base
- In both cases salt and water are formed
- Two of the most common amphoteric oxides are **zinc oxide**, **ZnO** and **aluminium oxide**, **Al_2O_3**
- The **hydroxides** of both of these elements also behave amphoterically
- Example of aluminium oxide behaving as a base:

$$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$$

- Example of aluminium oxide behaving as an acid:

$$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$$

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